

# ANURAG KHADELWAL

Associate Professor of Computer Science, Yale University

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## Work History

**Yale University, Department of Computer Science**

*Associate Professor, 2026–present*

*Assistant Professor, 2020–2026*

**Cornell University, Department of Computer Science (New York Campus)**

*Postdoctoral Associate, Fall 2019*

**Microsoft Research**

*Summer Research Intern, 2014 and 2012*

## Education

**University of California, Berkeley**

*PhD in Computer Science (2013–2019), Advised by Prof. Ion Stoica*

**Indian Institute of Technology, Kharagpur**

*B. Tech. in Computer Science and Engineering (2009–2013)*

## Research Focus

My research interests span computer software and hardware systems, networks, and security. My work addresses challenges in processing, storing, and serving large volumes of data to empower real-world systems: from sprawling internet services like social media to critical tools in health and medicine.

## Research Impact

[2025] [CORD](#) incorporated in the design of [NVIDIA's Vera Rubin architecture](#).

[2024] [PromptCache](#) adopted in [Gemini](#), [OpenAI](#), and [Anthropic](#) for reusing attention states across LLM prompts.

[2023] The ideas underlying [SCALO](#) have been tested by Yale Neurosurgery in FPGA-based emulation frameworks for interfacing with the human brain in the operating room.

[2021] [MIND](#) is the foundational system and central pillar of the multi-institution \$20-million NSF-funded AI Institute, [Athena](#).

[2019] [Confluo](#) employed in [Alibaba Cloud](#) for consuming and analyzing telemetry data.

[2015] [Succinct](#) employed by [Databricks customers](#) to query compressed RDDs.

## Awards & Honors

**Best Paper Award, ACM ASPLOS 2026** for *CounterPoint* [C4]

**Best Paper and Best Artifact Award Runner-Up, ACM MobiSys 2026** for *TimelyLLM* [C3]

**IEEE Micro Top Picks in Computer Architecture, 2025** for *CORD* [C9]

**Distinguished Artifact Award, ACM ISCA 2025** for *CORD* [C9]

**NetApp Faculty Fellowship, 2024** to develop GPU-native vector storage and indexing for AI

**Best Student Paper Award, ACM EuroSys 2024** for *TRINITY* [C13]

**IEEE Micro Top Picks in Computer Architecture, 2023** for *SCALO* [C15]

**Best Paper Award, ACM ISCA 2023** for *SCALO* [C15]

**Roberts Innovation Award, 2023** for innovations in data center resource disaggregation

**NSF CAREER Award, 2021** for in-network memory management for disaggregated data centers

**NetApp Faculty Fellowship, 2021** for elastic and compressed disaggregated memory at the edge

**Distinguished Paper Award, USENIX Security 2020** for *Pancake* [C21]

**Madan Sundar Sahu Scholarship, IIT Kharagpur, 2010 and 2011** for academic excellence

**Publications** Advisees (students and postdocs) are marked with <sup>†</sup>; \* denotes equal contribution.

CONFERENCE PAPERS

- [C1] [Efficient and Scalable Synchronization via Generalized Cache Coherence](#)  
Yanpeng Yu<sup>†</sup>, Seung-seob Lee<sup>†</sup>, Lin Zhong, and **Anurag Khandelwal**  
*USENIX OSDI*, 2026
- [C2] [BulletTime: Time Dilation for High-Fidelity Tracing](#)  
Michael Wu, Sibren Isaacman, Abhishek Bhattacharjee, and **Anurag Khandelwal**  
*ACM ISCA*, 2026
- [C3] [TimelyLLM: Time-sensitive LLM Serving System for Physical-I/O Limited Agents](#)  
Neiwen Ling, Guojun Chen, **Anurag Khandelwal**, and Lin Zhong  
*ACM MobiSys*, 2026  
**Best Paper Award Runner-Up, Best Artifact Award Runner-Up**
- [C4] [CounterPoint: Using Hardware Event Counters to Refute & Refine Microarchitectural Assumptions](#)  
Nick Lindsay, Caroline Trippel, **Anurag Khandelwal**, and Abhishek Bhattacharjee  
*ACM ASPLOS*, 2026  
**Best Paper Award**
- [C5] [Spirit: Fair Allocation of Interdependent Resources in Remote Memory Systems](#)  
Seung-seob Lee<sup>†</sup>, Jachym Putta<sup>†</sup>, Ziming Mao<sup>†</sup>, and **Anurag Khandelwal**  
*ACM SOSP*, 2025
- [C6] [Scalable Far Memory: Balancing Faults and Evictions](#)  
Yueyang Pan\*, Yash Lala\*<sup>†</sup>, Musa Unal, Yujie Ren, Seung-seob Lee<sup>†</sup>, Abhishek Bhattacharjee, **Anurag Khandelwal**, and Sanidhya Kashyap  
*ACM SOSP*, 2025
- [C7] [Found in Translation: A Generative Language Modeling Approach to Memory Access Pattern Attacks](#)  
Grace Jia<sup>†</sup>, Alex Wong, and **Anurag Khandelwal**  
*USENIX Security*, 2025
- [C8] [Weave: Efficient and Expressive Oblivious Analytics at Scale](#)  
Mahdi Soleimani<sup>†</sup>, Grace Jia<sup>†</sup>, and **Anurag Khandelwal**  
*USENIX OSDI*, 2025
- [C9] [CORD: Low-Latency, Bandwidth-Efficient and Scalable Release Consistency via Directory Ordering](#)  
Yanpeng Yu<sup>†</sup>, Nicolai Oswald, and **Anurag Khandelwal**  
*ACM ISCA*, 2025  
**Distinguished Artifact Award, IEEE Micro's Top Picks in Computer Architecture 2025**
- [C10] [PULSE: Accelerating Distributed Pointer-Traversals on Disaggregated Memory](#)  
Yupeng Tang<sup>†</sup>, Seung-seob Lee<sup>†</sup>, Abhishek Bhattacharjee, and **Anurag Khandelwal**  
*ACM ASPLOS*, 2025
- [C11] [Length Leakage in Oblivious Data Access Mechanisms](#)  
Grace Jia<sup>†</sup>, Rachit Agarwal, and **Anurag Khandelwal**  
*USENIX Security*, 2024
- [C12] [Prompt Cache: Modular attention reuse for low-latency inference](#)  
In Gim, Guojun Chen, Seung-seob Lee<sup>†</sup>, Nikhil Sarda, **Anurag Khandelwal**, and Lin Zhong  
*MLSys*, 2024
- [C13] [Trinity: A Fast Compressed Multi-attribute Data Store](#)  
Ziming Mao<sup>†</sup>, Kiran Srinivasan, and **Anurag Khandelwal**  
*ACM EuroSys*, 2024  
**Best Student Paper Award**
- [C14] [Karma: Resource Allocation for Dynamic Demands](#)  
Midhul Vuppapapati, Giannis Fikioris, Rachit Agarwal, Asaf Cidon, **Anurag Khandelwal**, and Éva Tardos  
*USENIX OSDI*, 2023

- [C15] [SCALO: An Accelerator-Rich Distributed System for Scalable Brain-Computer Interfacing](#)  
Karthik Sriram, Raghavendra Pothukuchi, Michał Gerasimiuk, Muhammed Ugur, Oliver Ye, Rajit Manohar, **Anurag Khandelwal**, and Abhishek Bhattacharjee  
*ACM ISCA*, 2023  
**Best Paper Award, IEEE Micro's Top Picks in Computer Architecture 2023**
- [C16] [Shepherd: Serving DNNs in the Wild](#)  
Hong Zhang, Yupeng Tang<sup>†</sup>, **Anurag Khandelwal**, and Ion Stoica  
*USENIX NSDI*, 2023
- [C17] [Shortstack: Distributed, Fault-tolerant, Oblivious Data Access](#)  
Midhul Vuppalapati\*, Kushal Babel\*, **Anurag Khandelwal**, and Rachit Agarwal  
*USENIX OSDI*, 2022
- [C18] [Jiffy: Elastic Far-memory for Stateful Serverless Analytics](#)  
**Anurag Khandelwal**, Yupeng Tang<sup>†</sup>, Rachit Agarwal, Aditya Akella, and Ion Stoica  
*ACM EuroSys*, 2022
- [C19] [MIND: In-Network Memory Management for Disaggregated Data Centers](#)  
Seung-seob Lee<sup>†</sup>, Yanpeng Yu<sup>†</sup>, Yupeng Tang<sup>†</sup>, **Anurag Khandelwal**, Lin Zhong, and Abhishek Bhattacharjee  
*ACM SOSP*, 2021
- [C20] [Caerus: Nimble Task Scheduling for Serverless Analytics](#)  
Hong Zhang, Yupeng Tang<sup>†</sup>, **Anurag Khandelwal**, Jingrong Chen, and Ion Stoica  
*USENIX NSDI*, 2021
- [C21] [Pancake: Frequency Smoothing for Encrypted Data Stores](#)  
Paul Grubbs\*, **Anurag Khandelwal**\*, Marie-Sarah Lacharité\*, Lloyd Brown, Lucy Li, Rachit Agarwal, and Thomas Ristenpart  
*USENIX Security*, 2020  
**Distinguished Paper Award**
- [C22] [Le Taureau: Deconstructing the Serverless Landscape & A Look Forward](#)  
**Anurag Khandelwal**, Arun Kejariwal, and Karthikeyan Ramasamy  
*ACM SIGMOD*, 2020
- [C23] [Confluo: Distributed Monitoring and Diagnosis Stack for High-speed Networks](#)  
**Anurag Khandelwal**, Rachit Agarwal, and Ion Stoica  
*USENIX NSDI*, 2019
- [C24] [ZipG: A Memory-efficient Graph Store for Interactive Queries](#)  
**Anurag Khandelwal**, Zongheng Yang, Evan Ye, Rachit Agarwal, and Ion Stoica  
*ACM SIGMOD*, 2017
- [C25] [BlowFish: Dynamic Storage-Performance Tradeoff in Data Stores](#)  
**Anurag Khandelwal**, Rachit Agarwal, and Ion Stoica  
*USENIX NSDI*, 2016
- [C26] [Succinct: Enabling Queries on Compressed Data](#)  
Rachit Agarwal, **Anurag Khandelwal**, and Ion Stoica  
*USENIX NSDI*, 2015

#### JOURNAL & MAGAZINE ARTICLES

- [J1] [Bringing Distributed Ordering to Heterogeneous Memory Consistency](#)  
Yanpeng Yu<sup>†</sup>, Nicolai Oswald, and **Anurag Khandelwal**  
*IEEE Micro*, 2026
- [J2] [Distributed Brain-Computer Interfacing With a Networked Multi-accelerator Architecture](#)  
Raghavendra Pothukuchi, Karthik Sriram, Michał Gerasimiuk, Muhammed Ugur, Rajit Manohar, **Anurag Khandelwal**, and Abhishek Bhattacharjee  
*IEEE Micro*, 2024
- [J3] [What Serverless Computing Is and Should Become: The Next Phase of Cloud Computing](#)  
Johann Schleier-Smith, Vikram Sreekanti, **Anurag Khandelwal**, Joao Carreira, Neeraja J. Yadwadkar, Raluca Ada Popa, Joseph E. Gonzalez, Ion Stoica, and David A. Patterson

*Communications of the ACM*, 2021

## WORKSHOP PAPERS

- [W1] [Exploring Intelligent Dynamic Resource Provisioning for Elastic Massive MIMO vRAN](#)  
Parthiban Annamalai, Minsung Kim, **Anurag Khandelwal**, and Lin Zhong  
*ACM HotMobile*, 2025
- [W2] [Prefetching Using Principles of Hippocampal-Neocortical Interaction](#)  
Michael Wu, Ketaki Joshi, Andrew Sheinberg, Guilherme Cox, **Anurag Khandelwal**, Raghavendra Pradyumna Pothukuchi, and Abhishek Bhattacharjee  
*ACM HotOS*, 2023

## DISSERTATION

- [T1] [Queries on Compressed Data](#)  
**Anurag Khandelwal**  
*PhD Dissertation, University of California, Berkeley*, 2019

## TECHNICAL REPORTS

- [R1] [Cloud Programming Simplified: A Berkeley View on Serverless Computing](#)  
Eric Jonas, Johann Schleier-Smith, Vikram Sreekanti, Chia-Che Tsai, **Anurag Khandelwal**, Qifan Pu, Vaishaal Shankar, João Carreira, Karl Krauth, Neeraja J. Yadwadkar, Joseph E. Gonzalez, Raluca Ada Popa, Ion Stoica, and David A. Patterson  
*UC Berkeley Technical Report*, 2019
- [R2] [Attacking Data Center Networks from the Inside](#)  
**Anurag Khandelwal**, Navendu Jain, and Seny Kamara  
*Microsoft Research Technical Report*, 2015

## PREPRINTS

- [P1] [Efficient MoE Serving in the Memory-Bound Regime: Balance Activated Experts, Not Tokens](#)  
Yanpeng Yu<sup>†</sup>, Haiyue Ma, Krish Agarwal, Nicolai Oswald, Qijing Huang, Hugo Linsenmaier, Chunhui Mei, Ritchie Zhao, Ritika Borkar, Bitu Darvish Rouhani, David Nellans, Ronny Krashinsky, and **Anurag Khandelwal**  
*arXiv Paper 2512/09277*, 2025
- [P2] [Wiretapping LLMs: Network Side-Channel Attacks on Interactive LLM Services](#)  
Mahdi Soleimani<sup>†</sup>, Grace Jia<sup>†</sup>, In Gim, Seung-seob Lee<sup>†</sup>, and **Anurag Khandelwal**  
*Cryptology ePrint Archive, Paper 2025/167*, 2025

## Funding

- [NSF SaTC: Fortifying and Enriching Confidential Computing Environments](#)  
2025–2028, \$900,000; PI: **Anurag Khandelwal**, Co-PIs: Lin Zhong, Seung-seob Lee
- [NetApp Faculty Fellowship: AI-Native Vector Storage and Indexing](#)  
2024–2025, \$117,686; PI: **Anurag Khandelwal**
- [Yale Engineering AI Seed Grant: Brain-Inspired Memory Systems for AI Infrastructure](#)  
2024–2025, \$150,000; PI: Abhishek Bhattacharjee, Co-PI: **Anurag Khandelwal**
- [Roberts Innovation Fund: Disaggregated Edge Clouds](#)  
2023–2024, \$50,000; PI: **Anurag Khandelwal**
- [NSF RINGS: Intelligent and Resilient Virtualization of Massive MIMO Physical Layer](#)  
2022–2025, \$1,000,000; PI: Lin Zhong, Co-PI: **Anurag Khandelwal**
- [Swebilius Foundation Award: Systems for Brain Data Acquisition and Analytics over Multiple Modalities and Time-Scales](#)  
2022, \$16,000; PI: Abhishek Bhattacharjee, Co-PIs: **Anurag Khandelwal**, Rajit Manohar

NSF AI Institute: Edge Computing Leveraging Next Generation Networks  
2021–2026, \$20,000,000; Multi-institution, multi-PI grant, \$575,000 to **Khandelwal**

NSF PPOSS Planning: High-Performance Certified Trust for Global-Scale Applications  
2021–2022, \$250,000; PI: Zhong Shao, Co-PIs: Abhishek Bhattacharjee, **Anurag Khandelwal**, Lin Zhong

NSF SaTC: Mixed Distribution Models for Encrypted Data Stores  
2021–2025, \$1,000,000; PI: **Anurag Khandelwal**, Co-PIs: Rachit Agarwal, Thomas Ristenpart

NetApp Faculty Fellowship: A Unified Data Stack for Next-Generation IoT Systems  
2021, \$50,000; PI: **Anurag Khandelwal**, Co-PI: Abhishek Bhattacharjee

NSF CAREER: In-Network Memory Management for Disaggregated Datacenters  
2021–2026, \$626,647; PI: **Anurag Khandelwal**

## Invited Talks

**Performant Shared Disaggregated Memory**  
Carnegie Mellon University (Nov 2025); University of Washington (Oct 2025); University of Texas at Austin (Oct 2025); Princeton University (Oct 2025); University of Pennsylvania (Oct 2025); Massachusetts Institute of Technology (Sep 2025); New York University (Mar 2025); Brown University (Mar 2025); NSF AI Institute Showcase, Duke University (Aug 2024); Microsoft Research (Aug 2024); AMD Research (Apr 2024)

**In-Network Memory Management for Disaggregated Data Centers**  
NSF AI Institute Showcase, Duke University (Aug 2022); Cornell University (Nov 2021); VMware (Nov 2021); Columbia University (Oct 2021)

**Jiffy: Ephemeral Storage for Serverless Analytics**  
Facebook (Mar 2021); Intel (Mar 2021)

**Serverless streaming architectures & algorithms for the enterprise**  
Global STAC Live (Oct 2020; industry conference for industry leaders in technology); Strata Data Conference (Sep 2019; industry conference of over 5000 companies)

**Building Interactive Query Systems**  
Yale University (Apr 2019); Purdue University (Mar 2019); Microsoft Research, Redmond (Mar 2019); Microsoft Research, Cambridge (UK) (Mar 2019)

**Realtime Monitoring and Analysis with Confluo**  
Splunk, San Francisco (Feb 2018)

**Succinct: Enabling Queries on Compressed Data**  
SF Big Analytics Meetup, San Francisco (Sep 2016); Bay Area AI Meetup, San Francisco (Sep 2016); Cloudera, San Francisco (Jun 2016)

## Teaching

*Course Director for all offerings below; YC enrollment in parentheses.*

**CPSC 422/522: Operating Systems** (24 lectures per term)  
Spring 2026 (36), Spring 2025 (34), Spring 2024 (44)

**CPSC 433/533: Computer Networks** (24 lectures per term)  
Spring 2022 (39), Spring 2021 (34), Spring 2020 (44)

**CPSC 438/538: Big Data Systems** (26 lectures per term)  
Fall 2025 (51), Fall 2023 (18), Fall 2021 (16)

**CPSC 638: Big Data Systems** (26 lectures per term)  
Fall 2020 (3)

## Advising

**Postdoctoral Researchers**  
Seung-seob Lee (2020–present, co-advised with Lin Zhong)

**Current PhD Students**  
Grace Jia (2022–present)

Mahdi Soleimani (2023–present)  
Yash Lala (2023–present)  
Stephen Chien (2026–present)

#### Graduated PhD Students

Yanpeng Yu (2021–2026) → Research Scientist at Databricks  
Yupeng Tang (2020–2024) → Research Scientist at Meta Research

#### Masters Students

Jachym Putta (2023–2025) → TechOps Engineer at DE Shaw  
Leo Li → PhD at Yale

#### Undergraduate Researchers

Rylan Yang (2025), Pax Ryan (2025), Rohan Thakur (2025–2026), Michael Tu (2024), Troy Feng (2023), Raymond Yang (2023), Ziming Mao (2021–2024) → PhD at UC Berkeley, Gabriel Buchdahl (2022–2023), Jachym Putta (2022–2023), Adit Gupta (2022–2023), Yuxuan Chen (2022), Eli Sage-Martinson (2022), Stanley Wong (2022), Daniel Li (2021–2022), Brian Liu (2021), Alan Chen (2021), Jonathan Kraft (2021), Eric Chang (2020), Jeff Ma → PhD at University of Michigan

## Education Outreach

#### Yale Pathways to Science, Summer Scholars Program

*“Through the Clouds: An Introduction to Cloud Computing”*

Instructor for 4-day workshop for high-school students (Summer 2021)

Instructor for 1-day seminar for high-school students (Summer 2022)

## Community Service

**Organization Committee:** CoNEXT’23 Poster Co-Chair

**Program Committee:** SIGCOMM 2020, NSDI 2020, NSDI 2021, NSDI 2022, HotNets 2022, NSDI 2023, EuroSys 2023, SOSP 2024, EuroSys 2025, OSDI 2025, NSDI 2026, SOSP 2026, EuroSys 2027

**Journal Editorial Board (Serverless Area):** JSys 2021

**External Program Committee:** ASPLOS 2021

**NSF Panels:** Undisclosed due to participant confidentiality.

## University Service

**Yale Quantitative Reasoning Council:** 2025–2026

**Yale College Dean’s Fellowship Review Committee:** 2026, 2024

## Department Service

**Chair, Colloquium Committee:** Fall 2021

**CS Undergraduate Advisor:** 2021–2022 (Freshman/Sophomore), 2024–2025 (Junior), 2025–2026 (Junior)

**Yale CS Research Internship Program:** Created and led the program (2021–2025)

**Faculty Hiring Committee:** 2025 (Database Systems Area Search)

**PhD Admissions Committee:** 2020, 2021, 2022, 2023, 2024, 2025, 2026

**Summer Systems Seminars:** 2020, 2021, 2022

**PhD Thesis Committee:** Bo Hu (2021), Karthik Sriram (2023), Yupeng Tang (2024), Yanpeng Yu (2026), Nick Lindsay (2026)